5

Topological dataset characteristics

Graph neural networks (GNNs) are designed to represent recommendation data as a bipartite, undirected graph. These graphs possess inherent topological properties that can and often are leveraged by GNN-based recommenders to enhance their representation learning capabilities.

This hypothesis is further supported by recent literature on representation learning that examines the role of graph structure (encompassing both topology and node features) in the performance of GNN models (Abu-El-Haija *et al.*, 2019; Castellana and Errica, 2023; Klicpera *et al.*, 2019; Luan *et al.*, 2019; Rong *et al.*, 2020; Wei *et al.*, 2022; Wu *et al.*, 2019; Xu *et al.*, 2018; Yang *et al.*, 2022; Zhao and Akoglu, 2020). Additionally, other studies have established connections between dataset characteristics and recommender system performance, reinforcing the plausibility of this relationship (Adomavicius and Zhang, 2012; Deldjoo *et al.*, 2020; Malitesta *et al.*, 2024).

This section provides a formal definition of four topological properties in graphs (Latapy *et al.*, 2008; Newman, 2003; Ravasz and Barabási, 2003): *node degree*, *clustering coefficient*, *degree assortativity*, and *degree distribution*. Then, it reinterprets these properties under the lens of recommender systems.

5.1 Node degree

Given a generic user u and item i, let $\mathcal{N}_u = \{i \in \mathcal{I} \mid R_{ui} = 1\}$ and $\mathcal{N}_i = \{u \in \mathcal{U} \mid R_{ui} = 1\}$ denote their respective neighborhood sets, as previously defined in Section 3.2.1. This definition naturally extends to l-hops, where $\mathcal{N}_u^{(l)}$ and $\mathcal{N}_i^{(l)}$ are the neighboring nodes of the user u and the item i, respectively, at l-hops distance. Then, the user and item **average node degrees** are defined as follows:

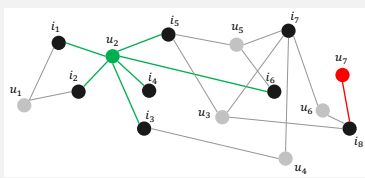
$$\sigma_{\mathcal{U}} = \frac{1}{U} \sum_{u \in \mathcal{U}} \sigma_u, \qquad \sigma_{\mathcal{I}} = \frac{1}{I} \sum_{i \in \mathcal{I}} \sigma_i, \tag{5.1}$$

where $\sigma_u = |\mathcal{N}_u^{(1)}|$ and $\sigma_i = |\mathcal{N}_i^{(1)}|$ denote the node degrees of user u and item i, respectively.

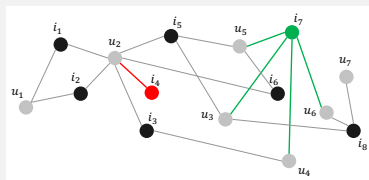
RecSys re-interpretation 1 (Node degree)**.** In the user-item graph, the node degree represents the number of items a user has interacted with or the number of users who have interacted with an item. It serves as a measure of user activity and item popularity within the dataset. A low node degree indicates a potential cold-start issue in recommendation, where cold users exhibit low engagement on the platform, and cold items correspond to niche products.

Node Degree in RecSys

A graphical representation of the node degree interpretation in the user-item graph. On the left, the user node degree is a measure of user activity. On the right, the item node degree distinguishes popular from niche items.



(a) **active** user *vs.* **inactive** user



(b) **popular** item *vs.* **niche** item

5.2 Clustering coefficient

For each partition in a bipartite graph, it is important to recognize nodes with overlapping neighborhoods. Regardless of their size, these overlaps reveal clusters of interconnected nodes. Let v and w be two nodes belonging to the same partition (e.g., user nodes). Their similarity can be measured using the intersection-over-union ratio of their neighborhoods (Latapy *et al.*, 2008). Formally, the clustering coefficient is defined as follows:

$$\gamma_v = \frac{\sum_{w \in \mathcal{N}_v^{(2)}} \gamma_{v,w}}{|\mathcal{N}_v^{(2)}|}, \qquad \text{with} \quad \gamma_{v,w} = \frac{|\mathcal{N}_v^{(1)} \cap \mathcal{N}_w^{(1)}|}{|\mathcal{N}_v^{(1)} \cup \mathcal{N}_w^{(1)}|}, \tag{5.2}$$

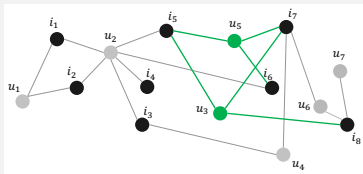
where, if $v, w \in \mathcal{U}$ are user nodes, we have $\mathcal{N}_v^{(2)} = \{w \in \mathcal{U} \mid \exists i \in \mathcal{I}, i \in \mathcal{N}_v \ \wedge \ i \in \mathcal{N}_w\}$ as the second-order neighborhood set of v. Considering second-order neighborhoods involves measuring similarities between nodes that are connected at a distance of (a multiple of) 2 hops. In a bipartite graph, this corresponds to identifying clusters of nodes within the same partition. Then, the **average clustering coefficient** on the node sets $\mathcal{U}$ and $\mathcal{I}$ is defined as follows:

$$\gamma_{\mathcal{U}} = \frac{1}{U} \sum_{u \in \mathcal{U}} \gamma_u, \qquad \gamma_{\mathcal{I}} = \frac{1}{I} \sum_{i \in \mathcal{I}} \gamma_i. \tag{5.3}$$

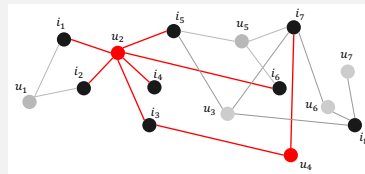
RecSys re-interpretation 2 (Clustering coefficient)**.** The clustering coefficient quantifies the extent to which a user (or item) node shares interactions with neighboring users (or items). High clustering coefficient values indicate a significant number of co-occurrences among nodes within the same partition. For instance, when considering the user-side formulation, the average clustering coefficient increases when multiple users interact with the same set of items. This intuition aligns with the core principle of collaborative filtering: users with overlapping interactions are likely to exhibit similar preferences.

Clustering Coefficient in RecSys

Graphical representation of the clustering coefficient interpretation in the user-item graph. On the left, an example illustrates two similar users sharing the 67% of the interacted items. On the right, two neighboring users exhibit different preferences.



(a) **similar** user activity ($\gamma_{u_3,u_5} = 0.667$)



(b) **different** user activity ($\gamma_{u_2,u_4} = 0.167$)

The clustering coefficient accounts for nodes at 2 distance hops. Nevertheless, we may want to capture properties for even more extended regions of the graph. For this reason, we introduce another topological characteristic that goes beyond the 2-hop distance.

5.3 Degree assortativity

In graph structures, nodes sharing similar properties show a tendency to group, and the assortativity quantifies this tendency. The definition of assortativity depends on the definition of similarity between nodes one decides to adopt. Therefore, diverse formulations of assortativity exist (Newman, 2003). In this work, we focus on the assortativity coefficient based on node degree similarities.

Let $\mathcal{D} = \{d_1, d_2, \dots\}$ represent the set of distinct node degree values in the graph, and let e_{d_h,d_k} denote the proportion of edges linking nodes with degrees d_h and d_k. We define q_{d_h} (q_{d_k}) as the probability of reaching a node with degree d_h (d_k) when traversing from another node with the same degree (i.e., the excess degree distribution). On such a basis, the **degree assortativity** is given by:

$$\rho = \frac{\sum_{d_h,d_k} d_h d_k (e_{d_h,d_k} - q_{d_h} q_{d_k})}{std_q^2}, \tag{5.4}$$

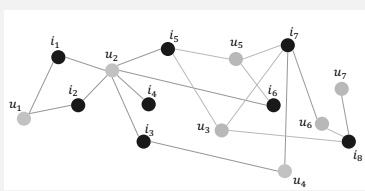
where std_q represents the standard deviation of the distribution q. Due to its formulation, the degree assortativity resembles a correlation measure, such as the Pearson correlation. Degree assortativity ranges from -1 to 1, where 1 indicates the highest similarity and -1 the lowest. When the value is below zero, we refer to it as *disassortativity*.

Similarly to the clustering coefficient, we aim to identify similarity patterns among nodes within the same partition. To achieve this, we first project the user-item bipartite graph into the user-user $\mathcal{G}_{\mathcal{U}}$ and the item-item graph $\mathcal{G}_{\mathcal{I}}$. We then compute the degree assortativity for $\mathcal{G}_{\mathcal{U}}$ and $\mathcal{G}_{\mathcal{I}}$, denoted as $\rho_{\mathcal{U}}$ and $\rho_{\mathcal{I}}$, respectively.

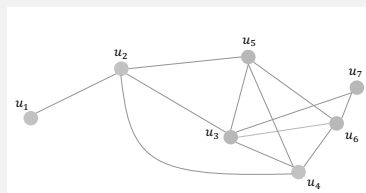
RecSys re-interpretation 3 (Degree assortativity)**.** The degree assortativity calculated user- and item-wise represents the tendency of users with the same activity level on the platform and items with the same popularity to gather, respectively. Since we calculate the degree assortativity on the complete user-user and item-item co-occurrence graphs, we deem this characteristic to provide a broader view of the dataset than the clustering coefficient. For this reason, we borrow the concept of search space traversal depth in search algorithms theory to give an intuition of degree assortativity. That is, we re-interpret degree assortativity in recommendation as a topological characteristic showing a strong look-ahead nature.

Degree Assortativity in RecSys

Graphical representation of the degree assortativity interpretation in the user-user graph. In this example, user nodes with the same degrees do not gather close in the user-user graph, thus resulting in a low value of degree assoratitvity.



(a) original user-item graph



(b) (**dis**)assortativity (ρ = -0.191)

5.4 Degree distribution

A network can be characterized through its degree distribution, defined as the probability distribution of all observed node degrees in the graph. Once again, let $\mathcal{D} = \{d_1, d_2, \dots\}$ be the set of distinct node degree values in the recommendation system, for both users and items. Then, let $n_d = |\{v \in \mathcal{U} \cup \mathcal{I} \mid \sigma_v = d\}|$ be the number of nodes (either user or item) that has degree d. The **degree distribution** of degree d is given by the following formula:

$$P(d) = \frac{n_d}{U + I}. \tag{5.5}$$

By considering the degree distribution of many real-world networks, it is commonly observed that they resemble the tendency of *scale-free* networks (Ravasz and Barabási, 2003): many nodes with very low degrees and few nodes with high degrees (also called *hubs*). The degree distribution of scale-free networks asymptotically follows the power law:

$$P(d) \sim d^{-\theta}, \text{ with } 2 < \theta < 3. \tag{5.6}$$

As observable in many recommendation datasets, the user-item graph follows this same tendency and can also be considered a *scale-free* network. This trend suggests some important properties of the user-item graph that can be re-interpreted through the lens of recommendation.

RecSys re-interpretation 4 (Degree distribution)**.** The *scale-free* tendency of the recommendation datasets implies some important properties. Many users and items exhibit very few interactions, while a few nodes (*hubs*) have numerous interactions, indicating they are highly active (in the case of users) and highly popular (in the case of items). As observed in other scale-free networks, this means that hubs will generally increase in their degree as the user-item graph evolves over time, leading to growing disparity between users and items (and sparsity of the dataset). All such properties are linked to the tendency of recommendation models to favor highly active users and highly popular items.

5.5 Empirical evaluation of topological characteristics

As previously done for the classical dataset characteristics (Section 4.5), here we analyze and discuss possible value trends in the topological dataset characteristics for our selected datasets. The computed values are reported in Table 5.1, where datasets are sorted by their number of interactions, from the largest (*MovieLens 1M*) to the smallest (*LastFM*) one.

In terms of **node degree** (σ), we observe almost aligned trends across all datasets, where the number of recorded interactions on the user and item side generally settles around 30-60. A slight difference is evident if considering the user and item levels separately, since the number of interactions on the item side are usually higher than on the user side. This is a clear indication of how items' popularity in the recommendation catalogs tend to be a prevalent phenomenon, representing a strong factor towards recommending most popular items (quite common for many recommendation models). The only deviation from the trend is represented by *MovieLens 1M* dataset, where the node degree is substantially higher than that of the other datasets. Indeed, as already evidenced for the classical dataset characteristics, this user-item graph appears to be extremely dense.

Regarding the **clustering coefficient** (γ), values are (again) quite aligned across datasets. Moreover, and unlike the node degree case, here we observe alignment also between the user and the item side, suggesting similar trends on the two levels. The highest values are measured for *MovieLens 1M* and *LastFM*, indicating that users (items) might have overlapping interactions in these two datasets. Put into other words: a certain group of users have interacted with the same items, and vice versa. We explain this trend by considering, once again, the density of these two datasets: indeed, the two user-item graphs show the most dense topological structures. Thus, it is reasonable to assume that providing high-quality recommendations in these two cases is easier than for the other datasets, actually and effectively exploiting the collaborative signal during the models' training.

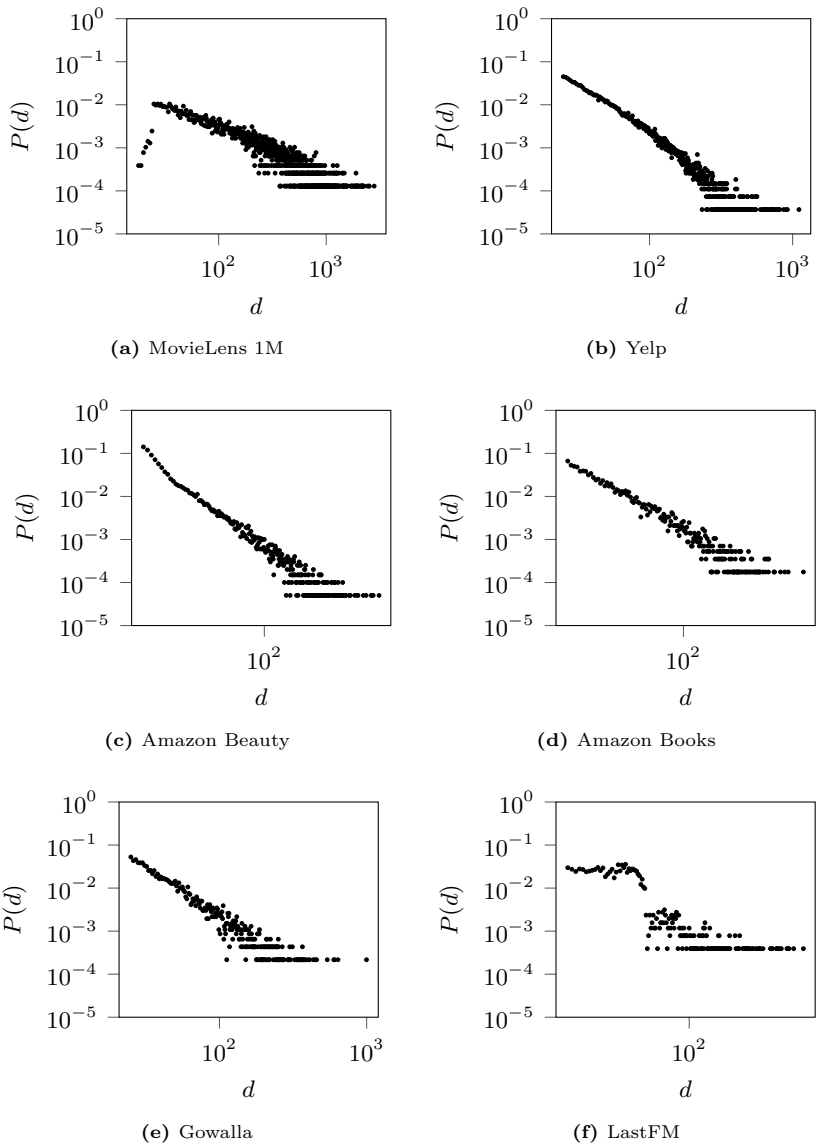
Some interesting insights are derived for the **degree assortativity** property (ρ). As observed, many computed values are negative, suggest-

Table 5.1: Calculated topological characteristics for our selected recommendation datasets.

Dataset	$\sigma_{\mathcal{U}}$	$\sigma_{\mathcal{I}}$	$\gamma_{\mathcal{U}}$	$\gamma_{\mathcal{I}}$	$\rho_{\mathcal{U}}$	$\rho_{\mathcal{I}}$
MovieLens 1M	175.317	335.506	0.065	0.055	-0.046	-0.040
Yelp	55.816	65.693	0.025	0.025	0.306	0.134
Amazon Beauty	36.130	26.449	0.021	0.025	-0.039	0.021
Amazon Books	43.486	57.686	0.033	0.026	0.031	-0.040
Gowalla	54.293	60.007	0.039	0.030	0.029	-0.029
LastFM	33.411	56.247	0.064	0.031	-0.029	-0.120

ing a dis-assortativity behavior. That is, users with the same activity levels do not tend to gather close in the user-user graph; the same can be affirmed for items and their popularity levels. A possible explanation for this phenomenon is as follows. When a new user enters the platform, she will likely tend to follow the preference patterns expressed by other highly-active users, which represent (with a certain degree of approximation) the average preferences in the whole community. Likewise, new items that are introduced into the platform will likely follow the interaction trends of already popular items in the system. The only exception to the rule is observed for *Yelp*, where the degree assortativity on both user and item sides is positive and strongly (if compared to the other values) assortative.

Finally, we plot the **degree distribution** of all the selected datasets in Figure 5.1. As already stated above, all degree distributions closely follow a power law tendency, suggesting a scale-free nature, as observed in most real-world social networks. Note that, only in this case, we do not compute and report any numerical value in Table 5.1 for this topological property because it is more effective to visualize it through its distributional shape (Figure 5.1). That is also why we do not associate this property with any symbol as previously done in Table 1.1.



(a) MovieLens 1M

(b) Yelp

(c) Amazon Beauty

(d) Amazon Books

(e) Gowalla

(f) LastFM

Figure 5.1: Computed degree distributions for all the selected datasets in this monograph. On the x and y axes, we report the degree values (both user and item side) and the probability distribution (in log scale), respectively.